Applied calculus questions

My role Item author and editor

I wrote application questions that connect calculus to real situations and show each step of the solution.

ORIGINAL DOCUMENT EXCERPT

Question:

A startup models its revenue as R(n) = n^2 million dollars when selling n thousand units. Use differentials to approximate the change in revenue when sales increase from n = 1.00 to n = 1.05 (in thousands).

Student Response box:

ΔR ≈ _____ (million dollars)

Answer:

0.10

Answer explanation:

Treat dR = R′(n) dn. Here R′(n) = 2n. At n = 1, R′(1) = 2. The change in n is dn = 0.05, so ΔR ≈ 2 × 0.05 = 0.10 million dollars.

Module URL and identifier withheld

Learning Objective Description: Investigate linear approximations and applications

Concept: Investigate linear approximations and applications

Atom Header: Text Instructional